

AMS 580 Quiz 1

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```
library(MASS)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.0
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## x dplyr::select() masks MASS::select()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(leaps)
```

```
## Warning: package 'leaps' was built under R version 4.3.3
```

```
library(car)
```

```
## Warning: package 'car' was built under R version 4.3.3
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some
```

```
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.3.3
```

```
## Loading required package: lattice
```

```
##
```

```
## Attaching package: 'caret'
```

```
##
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
## lift
```

```
library(dplyr)
```

Data Import

```
house <- read.table("/Users/thomashuang/Downloads/Ames_Housing_Data.csv", sep="," ,header=TRUE)
```

```
head(house)
```

```
##   Id LotArea OverallQual OverallCond YearBuilt YearRemodAdd CentralAir
## 1  1   8450         7         5      2003      2003           Y
## 2  2   9600         6         8      1976      1976           Y
## 3  3  11250         7         5      2001      2002           Y
## 4  4   9550         7         5      1915      1970           Y
## 5  5  14260         8         5      2000      2000           Y
## 6  6  14115         5         5      1993      1995           Y
##   X1stFlrSF X2ndFlrSF GrLivArea FullBath HalfBath BedroomAbvGr KitchenAbvGr
## 1      856      854     1710         2         1             3             1
## 2     1262         0     1262         2         0             3             1
## 3      920      866     1786         2         1             3             1
## 4      961      756     1717         1         0             3             1
## 5     1145     1053     2198         2         1             4             1
## 6      796      566     1362         1         1             1             1
##   TotRmsAbvGrd Fireplaces GarageCars GarageArea YrSold SalePrice
## 1             8           0           2        548   2008   208500
## 2             6           1           2        460   2007   181500
## 3             6           1           2        608   2008   223500
## 4             7           1           3        642   2006   140000
## 5             9           1           3        836   2008   250000
## 6             5           0           2        480   2009   143000
```

```
#remove Id column
```

```
house <- subset(house, select = -c(Id))
```

```
head(house)
```

```
##   LotArea OverallQual OverallCond YearBuilt YearRemodAdd CentralAir X1stFlrSF
## 1   8450         7         5      2003      2003           Y      856
```

```
## 2      9600          6          8      1976      1976          Y      1262
## 3     11250          7          5      2001      2002          Y      920
## 4      9550          7          5      1915      1970          Y      961
## 5     14260          8          5      2000      2000          Y     1145
## 6     14115          5          5      1993      1995          Y      796
##      X2ndFlrSF GrLivArea FullBath HalfBath BedroomAbvGr KitchenAbvGr TotRmsAbvGrd
## 1          854      1710          2          1          3          1          8
## 2           0      1262          2          0          3          1          6
## 3          866      1786          2          1          3          1          6
## 4          756      1717          1          0          3          1          7
## 5         1053      2198          2          1          4          1          9
## 6          566      1362          1          1          1          1          5
##      Fireplaces GarageCars GarageArea YrSold SalePrice
## 1           0          2          548    2008    208500
## 2           1          2          460    2007    181500
## 3           1          2          608    2008    223500
## 4           1          3          642    2006    140000
## 5           1          3          836    2008    250000
## 6           0          2          480    2009    143000
```

Data Split

```
set.seed(123)
training.samples <- house$SalePrice %>% createDataPartition(p=0.75,list = FALSE)
train.data <- house[training.samples, ]
test.data <- house[-training.samples, ]
```

Step-wise BIC

```
# null model
null <- lm(SalePrice ~ 1, data=train.data)

#full model
full <-lm(SalePrice ~. , data=train.data)
```

Forward

```
#find the best model in forward stepwise
fit_step_fwd_bic <- stepAIC(null, scope = list (lower = null, upper = full),
direction = "forward", k = log(nrow(train.data)), trace = 1)
```

```
## Start:  AIC=24805.6
## SalePrice ~ 1
##
##           Df Sum of Sq      RSS   AIC
## + OverallQual  1 4.5678e+12 2.6399e+12 23711
## + GrLivArea    1 3.9340e+12 3.2737e+12 23947
## + GarageCars   1 2.9822e+12 4.2255e+12 24227
```

```

## + GarageArea      1 2.8966e+12 4.3111e+12 24249
## + X1stFlrSF        1 2.8508e+12 4.3569e+12 24260
## + FullBath         1 2.3153e+12 4.8924e+12 24388
## + TotRmsAbvGrd    1 2.2046e+12 5.0031e+12 24412
## + YearBuilt        1 1.9200e+12 5.2877e+12 24473
## + YearRemodAdd     1 1.8773e+12 5.3304e+12 24482
## + Fireplaces       1 1.5633e+12 5.6444e+12 24544
## + X2ndFlrSF        1 7.3966e+11 6.4680e+12 24694
## + LotArea          1 5.9110e+11 6.6166e+12 24719
## + HalfBath         1 5.4828e+11 6.6594e+12 24726
## + CentralAir       1 4.3689e+11 6.7708e+12 24744
## + BedroomAbvGr    1 2.5739e+11 6.9503e+12 24773
## + KitchenAbvGr     1 1.3210e+11 7.0756e+12 24792
## <none>              7.2077e+12 24806
## + OverallCond      1 2.7572e+10 7.1801e+12 24808
## + YrSold            1 1.4182e+10 7.1935e+12 24810
##
## Step: AIC=23710.75
## SalePrice ~ OverallQual
##
##           Df Sum of Sq      RSS      AIC
## + GrLivArea    1 7.7987e+11 1.8600e+12 23334
## + X1stFlrSF    1 6.1545e+11 2.0244e+12 23426
## + TotRmsAbvGrd 1 3.8077e+11 2.2591e+12 23547
## + GarageArea   1 3.6672e+11 2.2731e+12 23554
## + LotArea      1 3.0237e+11 2.3375e+12 23584
## + GarageCars   1 2.9868e+11 2.3412e+12 23586
## + Fireplaces   1 1.9849e+11 2.4414e+12 23632
## + FullBath     1 1.5636e+11 2.4835e+12 23651
## + YearRemodAdd 1 5.9554e+10 2.5803e+12 23693
## + X2ndFlrSF    1 5.6070e+10 2.5838e+12 23694
## + YearBuilt     1 5.1999e+10 2.5879e+12 23696
## + BedroomAbvGr 1 5.0790e+10 2.5891e+12 23696
## + HalfBath      1 2.8967e+10 2.6109e+12 23706
## <none>          2.6399e+12 23711
## + CentralAir   1 4.9566e+09 2.6349e+12 23716
## + YrSold        1 2.7451e+09 2.6371e+12 23717
## + KitchenAbvGr 1 3.0071e+08 2.6396e+12 23718
## + OverallCond   1 1.6954e+08 2.6397e+12 23718
##
## Step: AIC=23333.63
## SalePrice ~ OverallQual + GrLivArea
##
##           Df Sum of Sq      RSS      AIC
## + X1stFlrSF    1 2.5018e+11 1.6098e+12 23182
## + X2ndFlrSF    1 2.2963e+11 1.6304e+12 23196
## + GarageArea   1 2.0246e+11 1.6575e+12 23214
## + YearBuilt     1 1.8228e+11 1.6777e+12 23228
## + GarageCars   1 1.4881e+11 1.7112e+12 23249
## + BedroomAbvGr 1 1.3008e+11 1.7299e+12 23261
## + LotArea      1 1.1039e+11 1.7496e+12 23274
## + YearRemodAdd 1 8.9774e+10 1.7702e+12 23286
## + KitchenAbvGr 1 3.9493e+10 1.8205e+12 23317
## + Fireplaces   1 3.3799e+10 1.8262e+12 23320

```

```

## + CentralAir      1 2.5828e+10 1.8342e+12 23325
## + TotRmsAbvGrd    1 1.8578e+10 1.8414e+12 23330
## + HalfBath        1 1.8117e+10 1.8419e+12 23330
## <none>              1.8600e+12 23334
## + YrSold           1 7.7611e+08 1.8592e+12 23340
## + FullBath         1 1.5251e+08 1.8598e+12 23340
## + OverallCond      1 1.0533e+08 1.8599e+12 23341
##
## Step:  AIC=23182.16
## SalePrice ~ OverallQual + GrLivArea + X1stFlrSF
##
##           Df Sum of Sq      RSS   AIC
## + YearBuilt    1 1.4114e+11 1.4687e+12 23088
## + GarageArea   1 1.1360e+11 1.4962e+12 23109
## + YearRemodAdd  1 8.9261e+10 1.5205e+12 23127
## + GarageCars   1 8.7310e+10 1.5225e+12 23128
## + BedroomAbvGr  1 8.4463e+10 1.5253e+12 23130
## + KitchenAbvGr  1 6.8749e+10 1.5411e+12 23141
## + LotArea       1 5.7740e+10 1.5521e+12 23149
## + CentralAir    1 1.6956e+10 1.5928e+12 23178
## + HalfBath      1 1.6947e+10 1.5929e+12 23178
## + TotRmsAbvGrd  1 1.4013e+10 1.5958e+12 23180
## + X2ndFlrSF     1 1.0678e+10 1.5991e+12 23182
## + Fireplaces    1 1.0413e+10 1.5994e+12 23182
## <none>              1.6098e+12 23182
## + OverallCond   1 5.1535e+09 1.6046e+12 23186
## + YrSold         1 1.1149e+09 1.6087e+12 23188
## + FullBath       1 2.1842e+08 1.6096e+12 23189
##
## Step:  AIC=23088.49
## SalePrice ~ OverallQual + GrLivArea + X1stFlrSF + YearBuilt
##
##           Df Sum of Sq      RSS   AIC
## + BedroomAbvGr  1 8.1383e+10 1.3873e+12 23033
## + KitchenAbvGr  1 6.5353e+10 1.4033e+12 23046
## + LotArea       1 6.1704e+10 1.4070e+12 23048
## + OverallCond   1 5.7453e+10 1.4112e+12 23052
## + GarageArea    1 5.6093e+10 1.4126e+12 23053
## + GarageCars    1 3.0749e+10 1.4379e+12 23072
## + FullBath      1 2.8789e+10 1.4399e+12 23074
## + YearRemodAdd  1 2.5236e+10 1.4434e+12 23076
## + Fireplaces    1 1.7214e+10 1.4514e+12 23083
## + TotRmsAbvGrd  1 1.0446e+10 1.4582e+12 23088
## <none>              1.4687e+12 23088
## + X2ndFlrSF     1 1.5606e+09 1.4671e+12 23094
## + HalfBath      1 7.6763e+08 1.4679e+12 23095
## + CentralAir    1 6.5649e+08 1.4680e+12 23095
## + YrSold         1 5.9223e+08 1.4681e+12 23095
##
## Step:  AIC=23032.96
## SalePrice ~ OverallQual + GrLivArea + X1stFlrSF + YearBuilt +
##           BedroomAbvGr
##           Df Sum of Sq      RSS   AIC

```

```

## + LotArea      1 6.2522e+10 1.3248e+12 22989
## + OverallCond  1 6.1741e+10 1.3255e+12 22990
## + KitchenAbvGr 1 5.6535e+10 1.3307e+12 22994
## + GarageArea   1 4.5453e+10 1.3418e+12 23003
## + GarageCars   1 2.3081e+10 1.3642e+12 23022
## + FullBath     1 1.6326e+10 1.3709e+12 23027
## + YearRemodAdd 1 1.5700e+10 1.3716e+12 23028
## + Fireplaces   1 1.0635e+10 1.3766e+12 23032
## <none>                1.3873e+12 23033
## + X2ndFlrSF    1 3.0503e+09 1.3842e+12 23038
## + CentralAir    1 1.7023e+09 1.3856e+12 23039
## + TotRmsAbvGrd 1 1.0514e+09 1.3862e+12 23039
## + YrSold        1 7.3566e+08 1.3865e+12 23039
## + HalfBath      1 4.2095e+07 1.3872e+12 23040
##
## Step: AIC=22989.37
## SalePrice ~ OverallQual + GrLivArea + X1stFlrSF + YearBuilt +
##      BedroomAbvGr + LotArea
##
##           Df Sum of Sq      RSS   AIC
## + OverallCond  1 5.4224e+10 1.2705e+12 22950
## + KitchenAbvGr 1 4.6223e+10 1.2785e+12 22957
## + GarageArea   1 3.7101e+10 1.2877e+12 22965
## + GarageCars   1 1.7448e+10 1.3073e+12 22982
## + YearRemodAdd 1 1.5016e+10 1.3097e+12 22984
## + FullBath     1 1.4826e+10 1.3099e+12 22984
## <none>                1.3248e+12 22989
## + Fireplaces   1 4.0513e+09 1.3207e+12 22993
## + X2ndFlrSF    1 2.2243e+09 1.3225e+12 22994
## + TotRmsAbvGrd 1 2.1955e+09 1.3226e+12 22994
## + CentralAir    1 6.6297e+08 1.3241e+12 22996
## + YrSold        1 3.9053e+08 1.3244e+12 22996
## + HalfBath      1 6.5241e+07 1.3247e+12 22996
##
## Step: AIC=22950.52
## SalePrice ~ OverallQual + GrLivArea + X1stFlrSF + YearBuilt +
##      BedroomAbvGr + LotArea + OverallCond
##
##           Df Sum of Sq      RSS   AIC
## + KitchenAbvGr 1 3.5189e+10 1.2353e+12 22927
## + GarageArea   1 3.5147e+10 1.2354e+12 22927
## + GarageCars   1 1.7736e+10 1.2528e+12 22942
## + FullBath     1 1.3148e+10 1.2574e+12 22946
## <none>                1.2705e+12 22950
## + Fireplaces   1 3.6154e+09 1.2669e+12 22954
## + TotRmsAbvGrd 1 3.2341e+09 1.2673e+12 22955
## + CentralAir    1 2.2357e+09 1.2683e+12 22956
## + X2ndFlrSF    1 1.4131e+09 1.2691e+12 22956
## + YearRemodAdd 1 1.3546e+09 1.2692e+12 22956
## + YrSold        1 1.0520e+09 1.2695e+12 22957
## + HalfBath      1 5.2399e+06 1.2705e+12 22958
##
## Step: AIC=22926.71
## SalePrice ~ OverallQual + GrLivArea + X1stFlrSF + YearBuilt +

```

```

##      BedroomAbvGr + LotArea + OverallCond + KitchenAbvGr
##
##      Df Sum of Sq      RSS      AIC
## + GarageArea      1 3.7118e+10 1.1982e+12 22900
## + GarageCars      1 2.1811e+10 1.2135e+12 22914
## + TotRmsAbvGrd    1 1.2850e+10 1.2225e+12 22922
## <none>                                1.2353e+12 22927
## + CentralAir      1 6.2453e+09 1.2291e+12 22928
## + FullBath        1 5.7701e+09 1.2296e+12 22929
## + X2ndFlrSF       1 2.6668e+09 1.2327e+12 22931
## + Fireplaces      1 1.2374e+09 1.2341e+12 22933
## + YearRemodAdd    1 1.1883e+09 1.2342e+12 22933
## + YrSold          1 6.9978e+08 1.2346e+12 22933
## + HalfBath        1 1.3264e+08 1.2352e+12 22934
##
## Step: AIC=22900.25
## SalePrice ~ OverallQual + GrLivArea + X1stFlrSF + YearBuilt +
##      BedroomAbvGr + LotArea + OverallCond + KitchenAbvGr + GarageArea
##
##      Df Sum of Sq      RSS      AIC
## + TotRmsAbvGrd    1 1.0361e+10 1.1879e+12 22898
## <none>                                1.1982e+12 22900
## + CentralAir      1 7.3618e+09 1.1909e+12 22900
## + FullBath        1 5.8701e+09 1.1924e+12 22902
## + X2ndFlrSF       1 2.9525e+09 1.1953e+12 22904
## + Fireplaces      1 1.8239e+09 1.1964e+12 22906
## + YearRemodAdd    1 1.4747e+09 1.1967e+12 22906
## + YrSold          1 5.2751e+08 1.1977e+12 22907
## + GarageCars      1 8.2145e+07 1.1981e+12 22907
## + HalfBath        1 2.8317e+07 1.1982e+12 22907
##
## Step: AIC=22897.72
## SalePrice ~ OverallQual + GrLivArea + X1stFlrSF + YearBuilt +
##      BedroomAbvGr + LotArea + OverallCond + KitchenAbvGr + GarageArea +
##      TotRmsAbvGrd
##
##      Df Sum of Sq      RSS      AIC
## <none>                                1.1879e+12 22898
## + CentralAir      1 7357491129 1.1805e+12 22898
## + FullBath        1 5585613921 1.1823e+12 22900
## + X2ndFlrSF       1 3305801334 1.1846e+12 22902
## + Fireplaces      1 1830748583 1.1860e+12 22903
## + YearRemodAdd    1 1011424162 1.1868e+12 22904
## + YrSold          1 539719160 1.1873e+12 22904
## + GarageCars      1 175129575 1.1877e+12 22905
## + HalfBath        1 51538397 1.1878e+12 22905

#display the BIC
BIC(fit_step_fwd_bic)

## [1] 26017.87

```

Backward

```
#find the best model in backward stepwise  
fit_step_back_bic <- stepAIC(full, k = log(nrow(train.data)), trace = 1,direction = "backward")
```

```
## Start:  AIC=22929.76  
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +  
##   YearRemodAdd + CentralAir + X1stFlrSF + X2ndFlrSF + GrLivArea +  
##   FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + TotRmsAbvGrd +  
##   Fireplaces + GarageCars + GarageArea + YrSold  
##  
##           Df  Sum of Sq      RSS   AIC  
## - GarageCars    1 2.0989e+08 1.1624e+12 22923  
## - YrSold         1 6.3120e+08 1.1628e+12 22923  
## - GrLivArea      1 1.8518e+09 1.1640e+12 22924  
## - HalfBath       1 2.0177e+09 1.1642e+12 22925  
## - YearRemodAdd   1 2.7209e+09 1.1649e+12 22925  
## - Fireplaces     1 3.1348e+09 1.1653e+12 22926  
## - X2ndFlrSF      1 3.7883e+09 1.1660e+12 22926  
## <none>           1.1622e+12 22930  
## - FullBath       1 9.8322e+09 1.1720e+12 22932  
## - TotRmsAbvGrd   1 9.8732e+09 1.1721e+12 22932  
## - CentralAir     1 9.9405e+09 1.1721e+12 22932  
## - X1stFlrSF      1 1.2202e+10 1.1744e+12 22934  
## - GarageArea     1 1.6150e+10 1.1783e+12 22938  
## - OverallCond    1 3.4890e+10 1.1971e+12 22955  
## - LotArea        1 3.6886e+10 1.1991e+12 22957  
## - KitchenAbvGr   1 3.7886e+10 1.2001e+12 22958  
## - BedroomAbvGr   1 5.9861e+10 1.2221e+12 22978  
## - YearBuilt      1 9.1829e+10 1.2540e+12 23006  
## - OverallQual    1 1.9540e+11 1.3576e+12 23093  
##  
## Step:  AIC=22922.95  
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +  
##   YearRemodAdd + CentralAir + X1stFlrSF + X2ndFlrSF + GrLivArea +  
##   FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + TotRmsAbvGrd +  
##   Fireplaces + GarageArea + YrSold  
##  
##           Df  Sum of Sq      RSS   AIC  
## - YrSold         1 6.0101e+08 1.1630e+12 22916  
## - GrLivArea      1 1.8958e+09 1.1643e+12 22918  
## - HalfBath       1 2.1119e+09 1.1645e+12 22918  
## - YearRemodAdd   1 2.6464e+09 1.1650e+12 22918  
## - Fireplaces     1 3.0215e+09 1.1654e+12 22919  
## - X2ndFlrSF      1 3.7410e+09 1.1661e+12 22920  
## <none>           1.1624e+12 22923  
## - TotRmsAbvGrd   1 9.7761e+09 1.1722e+12 22925  
## - CentralAir     1 9.9249e+09 1.1723e+12 22925  
## - FullBath       1 1.0152e+10 1.1726e+12 22926  
## - X1stFlrSF      1 1.2128e+10 1.1745e+12 22927  
## - OverallCond    1 3.5173e+10 1.1976e+12 22949  
## - GarageArea     1 3.6747e+10 1.1991e+12 22950  
## - LotArea        1 3.6788e+10 1.1992e+12 22950
```

```

## - KitchenAbvGr 1 3.8441e+10 1.2008e+12 22952
## - BedroomAbvGr 1 5.9675e+10 1.2221e+12 22971
## - YearBuilt 1 9.1673e+10 1.2541e+12 22999
## - OverallQual 1 1.9605e+11 1.3584e+12 23087
##
## Step: AIC=22916.52
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +
## YearRemodAdd + CentralAir + X1stFlrSF + X2ndFlrSF + GrLivArea +
## FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + TotRmsAbvGrd +
## Fireplaces + GarageArea
##
## Df Sum of Sq RSS AIC
## - GrLivArea 1 1.8774e+09 1.1649e+12 22911
## - HalfBath 1 2.1432e+09 1.1651e+12 22912
## - YearRemodAdd 1 2.4782e+09 1.1655e+12 22912
## - Fireplaces 1 2.9670e+09 1.1660e+12 22912
## - X2ndFlrSF 1 3.7890e+09 1.1668e+12 22913
## <none> 1.1630e+12 22916
## - TotRmsAbvGrd 1 9.7895e+09 1.1728e+12 22919
## - CentralAir 1 9.9174e+09 1.1729e+12 22919
## - FullBath 1 1.0227e+10 1.1732e+12 22919
## - X1stFlrSF 1 1.2189e+10 1.1752e+12 22921
## - OverallCond 1 3.4959e+10 1.1980e+12 22942
## - GarageArea 1 3.6900e+10 1.1999e+12 22944
## - LotArea 1 3.7099e+10 1.2001e+12 22944
## - KitchenAbvGr 1 3.8798e+10 1.2018e+12 22946
## - BedroomAbvGr 1 5.9664e+10 1.2227e+12 22964
## - YearBuilt 1 9.2247e+10 1.2552e+12 22993
## - OverallQual 1 1.9645e+11 1.3594e+12 23081
##
## Step: AIC=22911.29
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +
## YearRemodAdd + CentralAir + X1stFlrSF + X2ndFlrSF + FullBath +
## HalfBath + BedroomAbvGr + KitchenAbvGr + TotRmsAbvGrd + Fireplaces +
## GarageArea
##
## Df Sum of Sq RSS AIC
## - HalfBath 1 2.0115e+09 1.1669e+12 22906
## - YearRemodAdd 1 2.6644e+09 1.1675e+12 22907
## - Fireplaces 1 2.8099e+09 1.1677e+12 22907
## <none> 1.1649e+12 22911
## - CentralAir 1 9.8930e+09 1.1748e+12 22914
## - FullBath 1 9.9591e+09 1.1748e+12 22914
## - TotRmsAbvGrd 1 1.1301e+10 1.1762e+12 22915
## - OverallCond 1 3.3998e+10 1.1989e+12 22936
## - LotArea 1 3.7034e+10 1.2019e+12 22939
## - GarageArea 1 3.7453e+10 1.2023e+12 22939
## - KitchenAbvGr 1 4.0916e+10 1.2058e+12 22942
## - BedroomAbvGr 1 6.0331e+10 1.2252e+12 22960
## - YearBuilt 1 9.0557e+10 1.2554e+12 22986
## - X2ndFlrSF 1 1.8769e+11 1.3526e+12 23068
## - OverallQual 1 1.9854e+11 1.3634e+12 23077
## - X1stFlrSF 1 4.2093e+11 1.5858e+12 23243
##

```

```

## Step: AIC=22906.18
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +
##   YearRemodAdd + CentralAir + X1stFlrSF + X2ndFlrSF + FullBath +
##   BedroomAbvGr + KitchenAbvGr + TotRmsAbvGrd + Fireplaces +
##   GarageArea
##
##           Df Sum of Sq      RSS   AIC
## - Fireplaces    1 2.5070e+09 1.1694e+12 22902
## - YearRemodAdd  1 2.6377e+09 1.1695e+12 22902
## <none>                                1.1669e+12 22906
## - FullBath      1 8.0173e+09 1.1749e+12 22907
## - CentralAir    1 9.9056e+09 1.1768e+12 22908
## - TotRmsAbvGrd  1 1.1118e+10 1.1780e+12 22910
## - OverallCond   1 3.3894e+10 1.2008e+12 22931
## - LotArea       1 3.7537e+10 1.2044e+12 22934
## - GarageArea    1 3.7997e+10 1.2049e+12 22934
## - KitchenAbvGr  1 4.1028e+10 1.2079e+12 22937
## - BedroomAbvGr  1 5.9865e+10 1.2268e+12 22954
## - YearBuilt     1 9.1300e+10 1.2582e+12 22982
## - OverallQual   1 1.9928e+11 1.3662e+12 23072
## - X2ndFlrSF     1 2.2400e+11 1.3909e+12 23092
## - X1stFlrSF     1 4.1892e+11 1.5858e+12 23236
##
## Step: AIC=22901.54
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +
##   YearRemodAdd + CentralAir + X1stFlrSF + X2ndFlrSF + FullBath +
##   BedroomAbvGr + KitchenAbvGr + TotRmsAbvGrd + GarageArea
##
##           Df Sum of Sq      RSS   AIC
## - YearRemodAdd  1 1.9916e+09 1.1714e+12 22896
## <none>                                1.1694e+12 22902
## - FullBath      1 8.2979e+09 1.1777e+12 22902
## - CentralAir    1 8.9376e+09 1.1783e+12 22903
## - TotRmsAbvGrd  1 1.1181e+10 1.1806e+12 22905
## - OverallCond   1 3.4291e+10 1.2037e+12 22926
## - GarageArea    1 3.7167e+10 1.2066e+12 22929
## - LotArea       1 4.1366e+10 1.2108e+12 22933
## - KitchenAbvGr  1 4.3838e+10 1.2132e+12 22935
## - BedroomAbvGr  1 6.3470e+10 1.2329e+12 22952
## - YearBuilt     1 9.1019e+10 1.2604e+12 22977
## - OverallQual   1 2.0888e+11 1.3783e+12 23075
## - X2ndFlrSF     1 2.4529e+11 1.4147e+12 23103
## - X1stFlrSF     1 4.7434e+11 1.6437e+12 23268
##
## Step: AIC=22896.4
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +
##   CentralAir + X1stFlrSF + X2ndFlrSF + FullBath + BedroomAbvGr +
##   KitchenAbvGr + TotRmsAbvGrd + GarageArea
##
##           Df Sum of Sq      RSS   AIC
## - FullBath      1 7.4074e+09 1.1788e+12 22896
## <none>                                1.1714e+12 22896
## - CentralAir    1 9.0714e+09 1.1805e+12 22898
## - TotRmsAbvGrd  1 1.1966e+10 1.1834e+12 22901

```

```

## - GarageArea      1 3.6781e+10 1.2082e+12 22923
## - LotArea         1 4.1197e+10 1.2126e+12 22927
## - KitchenAbvGr    1 4.5133e+10 1.2165e+12 22931
## - OverallCond     1 4.7140e+10 1.2185e+12 22933
## - BedroomAbvGr    1 6.9403e+10 1.2408e+12 22952
## - YearBuilt       1 1.2725e+11 1.2986e+12 23002
## - OverallQual     1 2.1765e+11 1.3890e+12 23076
## - X2ndFlrSF       1 2.4686e+11 1.4183e+12 23099
## - X1stFlrSF       1 4.7547e+11 1.6469e+12 23263
##
## Step: AIC=22896.32
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +
##   CentralAir + X1stFlrSF + X2ndFlrSF + BedroomAbvGr + KitchenAbvGr +
##   TotRmsAbvGrd + GarageArea
##
##           Df Sum of Sq      RSS   AIC
## - CentralAir    1 7.3715e+09 1.1862e+12 22896
## <none>                                1.1788e+12 22896
## - TotRmsAbvGrd  1 1.2228e+10 1.1910e+12 22901
## - GarageArea    1 3.6460e+10 1.2153e+12 22923
## - LotArea       1 4.1286e+10 1.2201e+12 22927
## - OverallCond   1 4.6195e+10 1.2250e+12 22932
## - KitchenAbvGr  1 5.4264e+10 1.2331e+12 22939
## - BedroomAbvGr  1 7.7274e+10 1.2561e+12 22959
## - YearBuilt     1 1.2185e+11 1.3006e+12 22997
## - OverallQual   1 2.1115e+11 1.3900e+12 23070
## - X2ndFlrSF     1 2.4375e+11 1.4225e+12 23096
## - X1stFlrSF     1 4.7505e+11 1.6538e+12 23261
##
## Step: AIC=22896.15
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +
##   X1stFlrSF + X2ndFlrSF + BedroomAbvGr + KitchenAbvGr + TotRmsAbvGrd +
##   GarageArea
##
##           Df Sum of Sq      RSS   AIC
## <none>                                1.1862e+12 22896
## - TotRmsAbvGrd  1 1.2247e+10 1.1984e+12 22900
## - GarageArea    1 3.5374e+10 1.2215e+12 22921
## - OverallCond   1 3.9455e+10 1.2256e+12 22925
## - LotArea       1 4.0357e+10 1.2265e+12 22926
## - KitchenAbvGr  1 4.9364e+10 1.2355e+12 22934
## - BedroomAbvGr  1 8.0195e+10 1.2664e+12 22961
## - YearBuilt     1 1.1617e+11 1.3023e+12 22992
## - OverallQual   1 2.1235e+11 1.3985e+12 23070
## - X2ndFlrSF     1 2.4684e+11 1.4330e+12 23096
## - X1stFlrSF     1 4.7263e+11 1.6588e+12 23257

```

```
fit_step_back_bic
```

```

##
## Call:
## lm(formula = SalePrice ~ LotArea + OverallQual + OverallCond +
##   YearBuilt + X1stFlrSF + X2ndFlrSF + BedroomAbvGr + KitchenAbvGr +
##   TotRmsAbvGrd + GarageArea, data = train.data)

```

```
##
## Coefficients:
## (Intercept)      LotArea  OverallQual  OverallCond  YearBuilt
## -1.017e+06      7.340e-01    1.667e+04    5.943e+03    4.813e+02
##   X1stFlrSF      X2ndFlrSF  BedroomAbvGr  KitchenAbvGr  TotRmsAbvGrd
##   1.013e+02      6.478e+01   -1.488e+04   -3.343e+04    4.361e+03
##   GarageArea
##   3.560e+01
```

```
#display the BIC
BIC(fit_step_back_bic)
```

```
## [1] 26016.31
```

Both

```
#find the best model in backward stepwise
fit_step_both_bic <- stepAIC(full, k = log(nrow(train.data)), trace = 1,direction="both")
```

```
## Start:  AIC=22929.76
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +
##   YearRemodAdd + CentralAir + X1stFlrSF + X2ndFlrSF + GrLivArea +
##   FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + TotRmsAbvGrd +
##   Fireplaces + GarageCars + GarageArea + YrSold
##
##           Df Sum of Sq      RSS   AIC
## - GarageCars    1 2.0989e+08 1.1624e+12 22923
## - YrSold         1 6.3120e+08 1.1628e+12 22923
## - GrLivArea      1 1.8518e+09 1.1640e+12 22924
## - HalfBath       1 2.0177e+09 1.1642e+12 22925
## - YearRemodAdd   1 2.7209e+09 1.1649e+12 22925
## - Fireplaces     1 3.1348e+09 1.1653e+12 22926
## - X2ndFlrSF      1 3.7883e+09 1.1660e+12 22926
## <none>              1.1622e+12 22930
## - FullBath       1 9.8322e+09 1.1720e+12 22932
## - TotRmsAbvGrd   1 9.8732e+09 1.1721e+12 22932
## - CentralAir     1 9.9405e+09 1.1721e+12 22932
## - X1stFlrSF      1 1.2202e+10 1.1744e+12 22934
## - GarageArea     1 1.6150e+10 1.1783e+12 22938
## - OverallCond    1 3.4890e+10 1.1971e+12 22955
## - LotArea        1 3.6886e+10 1.1991e+12 22957
## - KitchenAbvGr   1 3.7886e+10 1.2001e+12 22958
## - BedroomAbvGr   1 5.9861e+10 1.2221e+12 22978
## - YearBuilt       1 9.1829e+10 1.2540e+12 23006
## - OverallQual    1 1.9540e+11 1.3576e+12 23093
##
## Step:  AIC=22922.95
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +
##   YearRemodAdd + CentralAir + X1stFlrSF + X2ndFlrSF + GrLivArea +
##   FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + TotRmsAbvGrd +
##   Fireplaces + GarageArea + YrSold
```

```

##
##           Df  Sum of Sq      RSS   AIC
## - YrSold      1 6.0101e+08 1.1630e+12 22916
## - GrLivArea    1 1.8958e+09 1.1643e+12 22918
## - HalfBath     1 2.1119e+09 1.1645e+12 22918
## - YearRemodAdd 1 2.6464e+09 1.1650e+12 22918
## - Fireplaces   1 3.0215e+09 1.1654e+12 22919
## - X2ndFlrSF    1 3.7410e+09 1.1661e+12 22920
## <none>                1.1624e+12 22923
## - TotRmsAbvGrd 1 9.7761e+09 1.1722e+12 22925
## - CentralAir   1 9.9249e+09 1.1723e+12 22925
## - FullBath     1 1.0152e+10 1.1726e+12 22926
## - X1stFlrSF    1 1.2128e+10 1.1745e+12 22927
## + GarageCars   1 2.0989e+08 1.1622e+12 22930
## - OverallCond  1 3.5173e+10 1.1976e+12 22949
## - GarageArea   1 3.6747e+10 1.1991e+12 22950
## - LotArea      1 3.6788e+10 1.1992e+12 22950
## - KitchenAbvGr 1 3.8441e+10 1.2008e+12 22952
## - BedroomAbvGr 1 5.9675e+10 1.2221e+12 22971
## - YearBuilt    1 9.1673e+10 1.2541e+12 22999
## - OverallQual  1 1.9605e+11 1.3584e+12 23087
##
## Step:  AIC=22916.52
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +
##   YearRemodAdd + CentralAir + X1stFlrSF + X2ndFlrSF + GrLivArea +
##   FullBath + HalfBath + BedroomAbvGr + KitchenAbvGr + TotRmsAbvGrd +
##   Fireplaces + GarageArea
##
##           Df  Sum of Sq      RSS   AIC
## - GrLivArea    1 1.8774e+09 1.1649e+12 22911
## - HalfBath     1 2.1432e+09 1.1651e+12 22912
## - YearRemodAdd 1 2.4782e+09 1.1655e+12 22912
## - Fireplaces   1 2.9670e+09 1.1660e+12 22912
## - X2ndFlrSF    1 3.7890e+09 1.1668e+12 22913
## <none>                1.1630e+12 22916
## - TotRmsAbvGrd 1 9.7895e+09 1.1728e+12 22919
## - CentralAir   1 9.9174e+09 1.1729e+12 22919
## - FullBath     1 1.0227e+10 1.1732e+12 22919
## - X1stFlrSF    1 1.2189e+10 1.1752e+12 22921
## + YrSold      1 6.0101e+08 1.1624e+12 22923
## + GarageCars   1 1.7970e+08 1.1628e+12 22923
## - OverallCond  1 3.4959e+10 1.1980e+12 22942
## - GarageArea   1 3.6900e+10 1.1999e+12 22944
## - LotArea      1 3.7099e+10 1.2001e+12 22944
## - KitchenAbvGr 1 3.8798e+10 1.2018e+12 22946
## - BedroomAbvGr 1 5.9664e+10 1.2227e+12 22964
## - YearBuilt    1 9.2247e+10 1.2552e+12 22993
## - OverallQual  1 1.9645e+11 1.3594e+12 23081
##
## Step:  AIC=22911.29
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +
##   YearRemodAdd + CentralAir + X1stFlrSF + X2ndFlrSF + FullBath +
##   HalfBath + BedroomAbvGr + KitchenAbvGr + TotRmsAbvGrd + Fireplaces +
##   GarageArea

```

```

##
##           Df Sum of Sq      RSS   AIC
## - HalfBath      1 2.0115e+09 1.1669e+12 22906
## - YearRemodAdd  1 2.6644e+09 1.1675e+12 22907
## - Fireplaces    1 2.8099e+09 1.1677e+12 22907
## <none>                                1.1649e+12 22911
## - CentralAir    1 9.8930e+09 1.1748e+12 22914
## - FullBath      1 9.9591e+09 1.1748e+12 22914
## - TotRmsAbvGrd  1 1.1301e+10 1.1762e+12 22915
## + GrLivArea     1 1.8774e+09 1.1630e+12 22916
## + YrSold        1 5.8256e+08 1.1643e+12 22918
## + GarageCars    1 2.2085e+08 1.1647e+12 22918
## - OverallCond   1 3.3998e+10 1.1989e+12 22936
## - LotArea       1 3.7034e+10 1.2019e+12 22939
## - GarageArea    1 3.7453e+10 1.2023e+12 22939
## - KitchenAbvGr  1 4.0916e+10 1.2058e+12 22942
## - BedroomAbvGr  1 6.0331e+10 1.2252e+12 22960
## - YearBuilt     1 9.0557e+10 1.2554e+12 22986
## - X2ndFlrSF     1 1.8769e+11 1.3526e+12 23068
## - OverallQual   1 1.9854e+11 1.3634e+12 23077
## - X1stFlrSF     1 4.2093e+11 1.5858e+12 23243
##
## Step:  AIC=22906.18
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +
##           YearRemodAdd + CentralAir + X1stFlrSF + X2ndFlrSF + FullBath +
##           BedroomAbvGr + KitchenAbvGr + TotRmsAbvGrd + Fireplaces +
##           GarageArea
##
##           Df Sum of Sq      RSS   AIC
## - Fireplaces    1 2.5070e+09 1.1694e+12 22902
## - YearRemodAdd  1 2.6377e+09 1.1695e+12 22902
## <none>                                1.1669e+12 22906
## - FullBath      1 8.0173e+09 1.1749e+12 22907
## - CentralAir    1 9.9056e+09 1.1768e+12 22908
## - TotRmsAbvGrd  1 1.1118e+10 1.1780e+12 22910
## + HalfBath      1 2.0115e+09 1.1649e+12 22911
## + GrLivArea     1 1.7456e+09 1.1651e+12 22912
## + YrSold        1 6.1313e+08 1.1663e+12 22913
## + GarageCars    1 3.1191e+08 1.1666e+12 22913
## - OverallCond   1 3.3894e+10 1.2008e+12 22931
## - LotArea       1 3.7537e+10 1.2044e+12 22934
## - GarageArea    1 3.7997e+10 1.2049e+12 22934
## - KitchenAbvGr  1 4.1028e+10 1.2079e+12 22937
## - BedroomAbvGr  1 5.9865e+10 1.2268e+12 22954
## - YearBuilt     1 9.1300e+10 1.2582e+12 22982
## - OverallQual   1 1.9928e+11 1.3662e+12 23072
## - X2ndFlrSF     1 2.2400e+11 1.3909e+12 23092
## - X1stFlrSF     1 4.1892e+11 1.5858e+12 23236
##
## Step:  AIC=22901.54
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +
##           YearRemodAdd + CentralAir + X1stFlrSF + X2ndFlrSF + FullBath +
##           BedroomAbvGr + KitchenAbvGr + TotRmsAbvGrd + GarageArea
##

```

```

##          Df Sum of Sq      RSS      AIC
## - YearRemodAdd  1 1.9916e+09 1.1714e+12 22896
## <none>                                1.1694e+12 22902
## - FullBath      1 8.2979e+09 1.1777e+12 22902
## - CentralAir    1 8.9376e+09 1.1783e+12 22903
## - TotRmsAbvGrd  1 1.1181e+10 1.1806e+12 22905
## + Fireplaces    1 2.5070e+09 1.1669e+12 22906
## + HalfBath      1 1.7085e+09 1.1677e+12 22907
## + GrLivArea     1 1.6117e+09 1.1678e+12 22907
## + YrSold        1 5.6072e+08 1.1688e+12 22908
## + GarageCars    1 1.7343e+08 1.1692e+12 22908
## - OverallCond   1 3.4291e+10 1.2037e+12 22926
## - GarageArea    1 3.7167e+10 1.2066e+12 22929
## - LotArea       1 4.1366e+10 1.2108e+12 22933
## - KitchenAbvGr  1 4.3838e+10 1.2132e+12 22935
## - BedroomAbvGr  1 6.3470e+10 1.2329e+12 22952
## - YearBuilt     1 9.1019e+10 1.2604e+12 22977
## - OverallQual   1 2.0888e+11 1.3783e+12 23075
## - X2ndFlrSF     1 2.4529e+11 1.4147e+12 23103
## - X1stFlrSF     1 4.7434e+11 1.6437e+12 23268
##
## Step: AIC=22896.4
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +
##           CentralAir + X1stFlrSF + X2ndFlrSF + FullBath + BedroomAbvGr +
##           KitchenAbvGr + TotRmsAbvGrd + GarageArea
##
##          Df Sum of Sq      RSS      AIC
## - FullBath      1 7.4074e+09 1.1788e+12 22896
## <none>                                1.1714e+12 22896
## - CentralAir    1 9.0714e+09 1.1805e+12 22898
## - TotRmsAbvGrd  1 1.1966e+10 1.1834e+12 22901
## + YearRemodAdd  1 1.9916e+09 1.1694e+12 22902
## + Fireplaces    1 1.8608e+09 1.1695e+12 22902
## + GrLivArea     1 1.7794e+09 1.1696e+12 22902
## + HalfBath      1 1.7234e+09 1.1697e+12 22902
## + YrSold        1 4.1749e+08 1.1710e+12 22903
## + GarageCars    1 1.3111e+08 1.1713e+12 22903
## - GarageArea    1 3.6781e+10 1.2082e+12 22923
## - LotArea       1 4.1197e+10 1.2126e+12 22927
## - KitchenAbvGr  1 4.5133e+10 1.2165e+12 22931
## - OverallCond   1 4.7140e+10 1.2185e+12 22933
## - BedroomAbvGr  1 6.9403e+10 1.2408e+12 22952
## - YearBuilt     1 1.2725e+11 1.2986e+12 23002
## - OverallQual   1 2.1765e+11 1.3890e+12 23076
## - X2ndFlrSF     1 2.4686e+11 1.4183e+12 23099
## - X1stFlrSF     1 4.7547e+11 1.6469e+12 23263
##
## Step: AIC=22896.32
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +
##           CentralAir + X1stFlrSF + X2ndFlrSF + BedroomAbvGr + KitchenAbvGr +
##           TotRmsAbvGrd + GarageArea
##
##          Df Sum of Sq      RSS      AIC
## - CentralAir    1 7.3715e+09 1.1862e+12 22896

```

```

## <none> 1.1788e+12 22896
## + FullBath 1 7.4074e+09 1.1714e+12 22896
## - TotRmsAbvGrd 1 1.2228e+10 1.1910e+12 22901
## + Fireplaces 1 2.2355e+09 1.1766e+12 22901
## + GrLivArea 1 1.5806e+09 1.1772e+12 22902
## + YearRemodAdd 1 1.1011e+09 1.1777e+12 22902
## + YrSold 1 4.8764e+08 1.1783e+12 22903
## + GarageCars 1 2.9586e+08 1.1785e+12 22903
## + HalfBath 1 3.7392e+07 1.1788e+12 22903
## - GarageArea 1 3.6460e+10 1.2153e+12 22923
## - LotArea 1 4.1286e+10 1.2201e+12 22927
## - OverallCond 1 4.6195e+10 1.2250e+12 22932
## - KitchenAbvGr 1 5.4264e+10 1.2331e+12 22939
## - BedroomAbvGr 1 7.7274e+10 1.2561e+12 22959
## - YearBuilt 1 1.2185e+11 1.3006e+12 22997
## - OverallQual 1 2.1115e+11 1.3900e+12 23070
## - X2ndFlrSF 1 2.4375e+11 1.4225e+12 23096
## - X1stFlrSF 1 4.7505e+11 1.6538e+12 23261
##
## Step: AIC=22896.15
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +
## X1stFlrSF + X2ndFlrSF + BedroomAbvGr + KitchenAbvGr + TotRmsAbvGrd +
## GarageArea
##
## Df Sum of Sq RSS AIC
## <none> 1.1862e+12 22896
## + CentralAir 1 7.3715e+09 1.1788e+12 22896
## + FullBath 1 5.7074e+09 1.1805e+12 22898
## - TotRmsAbvGrd 1 1.2247e+10 1.1984e+12 22900
## + GrLivArea 1 1.6128e+09 1.1846e+12 22902
## + Fireplaces 1 1.3516e+09 1.1848e+12 22902
## + YearRemodAdd 1 1.2806e+09 1.1849e+12 22902
## + YrSold 1 4.7839e+08 1.1857e+12 22903
## + GarageCars 1 2.8436e+08 1.1859e+12 22903
## + HalfBath 1 1.1220e+08 1.1861e+12 22903
## - GarageArea 1 3.5374e+10 1.2215e+12 22921
## - OverallCond 1 3.9455e+10 1.2256e+12 22925
## - LotArea 1 4.0357e+10 1.2265e+12 22926
## - KitchenAbvGr 1 4.9364e+10 1.2355e+12 22934
## - BedroomAbvGr 1 8.0195e+10 1.2664e+12 22961
## - YearBuilt 1 1.1617e+11 1.3023e+12 22992
## - OverallQual 1 2.1235e+11 1.3985e+12 23070
## - X2ndFlrSF 1 2.4684e+11 1.4330e+12 23096
## - X1stFlrSF 1 4.7263e+11 1.6588e+12 23257

```

```
fit_step_both_bic
```

```

##
## Call:
## lm(formula = SalePrice ~ LotArea + OverallQual + OverallCond +
## YearBuilt + X1stFlrSF + X2ndFlrSF + BedroomAbvGr + KitchenAbvGr +
## TotRmsAbvGrd + GarageArea, data = train.data)
##
## Coefficients:

```

```
## (Intercept)      LotArea OverallQual OverallCond   YearBuilt
## -1.017e+06      7.340e-01   1.667e+04   5.943e+03   4.813e+02
##   X1stFlrSF      X2ndFlrSF BedroomAbvGr KitchenAbvGr TotRmsAbvGrd
##   1.013e+02      6.478e+01   -1.488e+04   -3.343e+04   4.361e+03
##   GarageArea
##   3.560e+01
```

```
#display the BIC
BIC(fit_step_both_bic)
```

```
## [1] 26016.31
```

Step-wise Result

Since Backward yield the smallest BIC.

Coefficients

```
summary(fit_step_back_bic)$coefficients[,1]
```

```
## (Intercept)      LotArea OverallQual OverallCond   YearBuilt
## -1.017464e+06  7.339645e-01  1.667461e+04  5.943478e+03  4.812917e+02
##   X1stFlrSF      X2ndFlrSF BedroomAbvGr KitchenAbvGr TotRmsAbvGrd
##   1.013470e+02  6.477953e+01 -1.488137e+04 -3.342536e+04  4.361293e+03
##   GarageArea
##   3.559750e+01
```

RMSE and R²

```
pred_step_back_bic <- fit_step_back_bic %>% predict(test.data)
data_frame(
  RMSE = RMSE(pred_step_back_bic, test.data$SalePrice),
  Rsquare = R2(pred_step_back_bic, test.data$SalePrice)
)
```

```
## Warning: 'data_frame()' was deprecated in tibble 1.1.0.
## i Please use 'tibble()' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
## # A tibble: 1 x 2
##   RMSE Rsquare
##   <dbl> <dbl>
## 1 48728. 0.660
```

Best Subset SSE

```

# use best subset to find the model
fit_bs <- regsubsets(SalePrice ~., data=train.data, nvmax = 30)

#save summary
result <- summary(fit_bs)

#display the sse of each model candidate
result$rss

## [1] 2.639851e+12 1.859983e+12 1.599657e+12 1.468656e+12 1.387273e+12
## [6] 1.324751e+12 1.270527e+12 1.235339e+12 1.198221e+12 1.186167e+12
## [11] 1.178795e+12 1.171388e+12 1.169396e+12 1.166889e+12 1.164878e+12
## [16] 1.163001e+12 1.162400e+12 1.162190e+12

#find which model that have the least sse
which.min(result$rss)

## [1] 18

#sse of the 18th model
result$rss[18]

## [1] 1.16219e+12

```

From this, we know the best model based on the SSE criterion is the model with 18 predictors.

```

result$which[18,]

## (Intercept)      LotArea OverallQual OverallCond   YearBuilt YearRemodAdd
##          TRUE         TRUE         TRUE         TRUE         TRUE         TRUE
## CentralAirY    X1stFlrSF    X2ndFlrSF    GrLivArea    FullBath    HalfBath
##          TRUE         TRUE         TRUE         TRUE         TRUE         TRUE
## BedroomAbvGr KitchenAbvGr TotRmsAbvGrd  Fireplaces  GarageCars  GarageArea
##          TRUE         TRUE         TRUE         TRUE         TRUE         TRUE
##          YrSold
##          TRUE

```

The result above indicate that the full model has the lowest SSE of 1.16219e+12.

Coefficients

```

#fit the full model
fit_full <- lm(SalePrice ~. , data=train.data)

#the coefficient of full model
summary(fit_full)$coefficients[,1]

```

```
## (Intercept)      LotArea OverallQual OverallCond      YearBuilt
## -2.629414e+05  7.105257e-01  1.649082e+04  6.308172e+03  5.862392e+02
## YearRemodAdd    CentralAirY    X1stFlrSF    X2ndFlrSF    GrLivArea
##  1.117984e+02 -1.419098e+04  7.460499e+01  4.150130e+01  2.886863e+01
##      FullBath      HalfBath  BedroomAbvGr  KitchenAbvGr  TotRmsAbvGrd
## -9.006764e+03 -3.904000e+03 -1.329971e+04 -3.067973e+04  3.972705e+03
##      Fireplaces  GarageCars  GarageArea      YrSold
##  3.233073e+03 -1.360815e+03  4.003564e+01 -5.808511e+02
```

RMSE and R^2

```
pred_full <- fit_full %>% predict(test.data)
data_frame(
  RMSE = RMSE(pred_full, test.data$SalePrice),
  Rsquare = R2(pred_full, test.data$SalePrice)
)
```

```
## # A tibble: 1 x 2
##   RMSE Rsquare
##   <dbl>   <dbl>
## 1 49485.   0.655
```

Which is the best?

Which model selection method among the 2 we have used above is the best?

Step-wise selection method performs the best.

- (a) Please compare the BIC of these models using the training data, as well as display these two models so we can see the parameter estimators and model goodness of fit measures.

```
#Stepwise best BIC
BIC(fit_step_back_bic)
```

```
## [1] 26016.31
```

```
#Stepwise best summary
summary(fit_step_back_bic)
```

```
##
## Call:
## lm(formula = SalePrice ~ LotArea + OverallQual + OverallCond +
##      YearBuilt + X1stFlrSF + X2ndFlrSF + BedroomAbvGr + KitchenAbvGr +
##      TotRmsAbvGrd + GarageArea, data = train.data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -124802  -17854   -1674   15227  245317
##
## Coefficients:
```

```
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.017e+06  9.199e+04 -11.060 < 2e-16 ***
## LotArea      7.340e-01  1.207e-01   6.079 1.68e-09 ***
## OverallQual   1.667e+04  1.196e+03  13.943 < 2e-16 ***
## OverallCond   5.943e+03  9.889e+02   6.010 2.53e-09 ***
## YearBuilt     4.813e+02  4.667e+01  10.313 < 2e-16 ***
## X1stFlrSF     1.013e+02  4.872e+00  20.802 < 2e-16 ***
## X2ndFlrSF     6.478e+01  4.309e+00  15.033 < 2e-16 ***
## BedroomAbvGr -1.488e+04  1.737e+03  -8.569 < 2e-16 ***
## KitchenAbvGr -3.343e+04  4.972e+03  -6.723 2.88e-11 ***
## TotRmsAbvGrd  4.361e+03  1.302e+03   3.349 0.00084 ***
## GarageArea    3.560e+01  6.255e+00   5.691 1.62e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 33050 on 1086 degrees of freedom
## Multiple R-squared:  0.8354, Adjusted R-squared:  0.8339
## F-statistic: 551.3 on 10 and 1086 DF, p-value: < 2.2e-16
```

```
#best subset best BIC
BIC(full)
```

```
## [1] 26049.91
```

```
#best subset best
summary(full)
```

```
##
## Call:
## lm(formula = SalePrice ~ ., data = train.data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -121064  -17483   -2009    15166   242604
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.629e+05  1.524e+06  -0.173 0.863031
## LotArea      7.105e-01  1.215e-01   5.849 6.54e-09 ***
## OverallQual   1.649e+04  1.225e+03  13.463 < 2e-16 ***
## OverallCond   6.308e+03  1.109e+03   5.689 1.65e-08 ***
## YearBuilt     5.862e+02  6.352e+01   9.229 < 2e-16 ***
## YearRemodAdd  1.118e+02  7.037e+01   1.589 0.112437
## CentralAirY   -1.419e+04  4.673e+03  -3.037 0.002451 **
## X1stFlrSF     7.461e+01  2.218e+01   3.364 0.000795 ***
## X2ndFlrSF     4.150e+01  2.214e+01   1.875 0.061127 .
## GrLivArea     2.887e+01  2.203e+01   1.311 0.190275
## FullBath      -9.007e+03  2.982e+03  -3.020 0.002588 **
## HalfBath      -3.904e+03  2.854e+03  -1.368 0.171586
## BedroomAbvGr -1.330e+04  1.785e+03  -7.451 1.89e-13 ***
## KitchenAbvGr -3.068e+04  5.175e+03  -5.928 4.12e-09 ***
## TotRmsAbvGrd  3.973e+03  1.313e+03   3.026 0.002535 **
## Fireplaces    3.233e+03  1.896e+03   1.705 0.088446 .
```

```
## GarageCars    -1.361e+03  3.084e+03  -0.441 0.659136
## GarageArea     4.004e+01  1.034e+01   3.870 0.000115 ***
## YrSold         -5.809e+02  7.591e+02  -0.765 0.444341
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 32830 on 1078 degrees of freedom
## Multiple R-squared:  0.8388, Adjusted R-squared:  0.8361
## F-statistic: 311.5 on 18 and 1078 DF,  p-value: < 2.2e-16
```

(b) Furthermore, please compare the RMSE and R2 of these models using the test data.

```
# RMSE and R2 of step-wise
pred_step_back_bic <- fit_step_back_bic %>% predict(test.data)
data_frame(
  RMSE = RMSE(pred_step_back_bic, test.data$SalePrice),
  Rsquare = R2(pred_step_back_bic, test.data$SalePrice)
)
```

```
## # A tibble: 1 x 2
##   RMSE Rsquare
##   <dbl>   <dbl>
## 1 48728.   0.660
```

```
# RMSE and R2 of best subset
pred_full <- fit_full %>% predict(test.data)
data_frame(
  RMSE = RMSE(pred_full, test.data$SalePrice),
  Rsquare = R2(pred_full, test.data$SalePrice)
)
```

```
## # A tibble: 1 x 2
##   RMSE Rsquare
##   <dbl>   <dbl>
## 1 49485.   0.655
```

(c) Please discuss any modifications you can do to further improve your model(s).

For improvement, I won't use SSE criterion for model selection since full model generally has the least SSE which doesn't account for model complexity.